

Homework 5

Solution

1. Compute each of the following improper integrals or prove that it doesn't converge

(a) $\int_{-\infty}^{\infty} e^{-|x|} dx$.

Solution:

We have

$$\begin{aligned} \int_{-\infty}^{\infty} e^{-|x|} dx &= \int_{-\infty}^0 e^{-|x|} dx + \int_0^{\infty} e^{-|x|} dx = \int_{-\infty}^0 e^x dx + \int_0^{\infty} e^{-x} dx = [e^x]_{-\infty}^0 + [-e^{-x}]_0^{\infty} \\ &= 1 - e^{-\infty} - e^{-\infty} + e^0 = 2 \end{aligned}$$

(b) $\int_{-\infty}^{\infty} \frac{x}{\pi(1+x^2)} dx$ (we will see this integral in the course Probability)

Solution:

We note that

$$\int_0^{\infty} \frac{x}{\pi(1+x^2)} dx = \lim_{x \rightarrow \infty} \int_0^x \frac{t}{\pi(1+t^2)} dt = \lim_{x \rightarrow \infty} \left[\frac{1}{\pi} \frac{\ln|1+t^2|}{2} \right]_0^x = \lim_{x \rightarrow \infty} \left[\frac{1}{\pi} \left(\frac{\ln|1+x^2|}{2} - \frac{\ln|1+0|}{2} \right) \right] = \infty$$

Therefore the integral $\int_{-\infty}^{\infty} \frac{x}{\pi(1+x^2)} dx$ diverges.

(c) $\int_0^{\infty} x^2 \cdot \lambda e^{-\lambda x} dx$, where $\lambda > 0$ (we will see this integral in the course Probability).

Solution:

By HW3Q1(c) we know that

$$\int x^2 \cdot \lambda e^{-\lambda x} dx = -\frac{e^{-\lambda x} (\lambda^2 x^2 + 2x\lambda + 2)}{\lambda^2} + C$$

Thus

$$\begin{aligned} \int_0^{\infty} [x^2 \cdot \lambda e^{-\lambda x}] dx &= \left[-\frac{e^{-\lambda x} (\lambda^2 x^2 + 2x\lambda + 2)}{\lambda^2} \right]_0^{\infty} \\ &= \lim_{x \rightarrow \infty} \left[-\frac{e^{-\lambda x} (\lambda^2 x^2 + 2x\lambda + 2)}{\lambda^2} - \left(-\frac{e^{-\lambda \cdot 0} (\lambda^2 \cdot 0^2 + 2 \cdot 0 \cdot \lambda + 2)}{\lambda^2} \right) \right] \\ &= \lim_{x \rightarrow \infty} \left(-\frac{1}{\lambda^2} \cdot \frac{\lambda^2 x^2 + 2x\lambda + 2}{e^{\lambda x}} + \frac{2}{\lambda^2} \right) \stackrel{(\star\star)}{=} -\frac{1}{\lambda^2} \cdot 0 + \frac{2}{\lambda^2} \\ &= \frac{2}{\lambda^2} \end{aligned}$$

where

$$\lim_{x \rightarrow \infty} \frac{\lambda^2 x^2 + 2x\lambda + 2}{e^{\lambda x}} \stackrel{\frac{\infty}{\infty}}{=} \lim_{x \rightarrow \infty} \frac{2\lambda^2 x + 2\lambda}{\lambda \cdot e^{\lambda x}} \stackrel{\frac{\infty}{\infty}}{=} \lim_{x \rightarrow \infty} \frac{2\lambda^2}{\lambda^2 \cdot e^{\lambda x}} \stackrel{\text{AOL}}{=} 0 \quad (\star\star)$$

2. Determine whether each of the following improper integrals converges or diverges

(a) $\int_1^\infty e^{-x} \ln(x) dx$.

Solution:

First, we clearly have $e^{-x} \ln(x) \geq 0$ for every $x \geq 1$. Thus we can use the comparison test. Note that

$$\lim_{x \rightarrow \infty} x e^{-x} = \lim_{x \rightarrow \infty} \frac{x}{e^x} \stackrel{\frac{\infty}{\infty}}{=} \lim_{x \rightarrow \infty} \frac{1}{e^x} = 0 \quad (\star)$$

We know that $\ln(1+x) \leq x$ for every $x > -1$. Thus, $\ln(x) \leq x - 1 \leq x$ for every $x \geq 1$. We now compute

$$\int_1^\infty e^{-x} \ln(x) dx \leq \int_1^\infty x e^{-x} dx \stackrel{\text{IBP}}{=} [-x e^{-x}]_1^\infty + \int_1^\infty e^{-x} dx \stackrel{(\star)}{=} \frac{1}{e} + [-e^{-x}]_1^\infty = \frac{2}{e}$$

Thus, the integral converges by the comparison test.

(b) $\int_1^\infty \frac{7x + \sin x}{2x^3 + x + 1} dx$.

Solution:

First, we clearly have $\frac{7x + \sin x}{2x^3 + x + 1} \geq 0$ for every $x \geq 1$. Thus we can use the comparison test. We note that

$$\int_1^\infty \frac{7x + \sin x}{2x^3 + x + 1} dx \leq \int_1^\infty \frac{7x + x}{2x^3} dx = \int_1^\infty \frac{8x}{2x^3} dx = \int_1^\infty \frac{4}{x^2} dx = 4 \int_1^\infty \frac{dx}{x^2} < \infty$$

Thus, by the comparison test $\int_1^\infty \frac{7x + \sin x}{2x^3 + x + 1} dx < \infty$.

(c) $\int_1^3 \frac{x}{\ln(x)} dx$.

Solution:

First, we clearly have $\frac{x}{\ln(x)} \geq 0$ for every $1 < x \leq 3$. Thus we can use the comparison test. We proved that for every $x > 1$ we have $\ln(1+x) \leq x$, thus

$$\ln(x) \leq x - 1 \Rightarrow \frac{1}{\ln x} \geq \frac{1}{x - 1}$$

and therefore

$$\int_1^3 \frac{x}{\ln(x)} dx \geq \int_1^3 \frac{x}{x-1} dx \geq \int_1^3 \frac{dx}{x-1} \stackrel{t=x-1, dt=dx}{=} \int_0^2 \frac{dt}{t} \geq \int_0^1 \frac{dt}{t} = \infty$$

Thus by the comparison test

$$\int_1^3 \frac{x}{\ln(x)} dx = \infty$$

3. Let f be a function that is continuous on $[0, 2]$. Prove there exists a point $c \in [0, 2]$ such that

$$\int_0^2 x^2 f(x) dx = \frac{8 \cdot f(c)}{3}$$

Solution:

First, as f is continuous on $[0, 2]$, then f is integrable on $[0, 2]$ and therefore $\int_0^2 x^2 f(x) dx$ is well-defined.

As f is continuous on $[0, 2]$, then by Weierstrass's theorem there exist $x_1, x_2 \in [0, 2]$ such that

$$f(x_2) \leq f(x) \leq f(x_1) \quad \forall x \in [0, 2]$$

We denote $f(x_1) = M$ and $f(x_2) = m$ and therefore

$$m \leq f(x) \leq M \quad \forall x \in [0, 2]$$

and therefore

$$mx^2 \leq x^2 f(x) \leq Mx^2 \quad \forall x \in [0, 2]$$

Choose $r = \frac{3}{8} \int_0^2 x^2 f(x) dx$. By the monotonicity of the definite integral,

$$r = \frac{3}{8} \int_0^2 x^2 f(x) dx \leq \frac{3}{8} \int_0^2 Mx^2 dx = \frac{3M}{8} \int_0^2 x^2 dx = \frac{3M}{8} \cdot \left[\frac{x^3}{3} \right]_0^2 = \frac{3M}{8} \cdot \frac{8}{3} = M$$

and similarly, $r \geq m$. It follows that $m \leq r \leq M$.

If $x_1 = x_2$, then f is constant and therefore for every $c \in [0, 2]$ we have $f(c) = r = m = M \Rightarrow \int_0^2 x^2 f(x) = \frac{8 \cdot f(c)}{3}$.

Otherwise, as f is continuous, then by the generalized IVT there exists a c between x_1 and x_2 (namely, $c \in [x_1, x_2] \subseteq [0, 2]$ or $c \in [x_2, x_1] \subseteq [0, 2]$) such that $f(c) = r \Rightarrow \int_0^2 x^2 f(x) = \frac{8 \cdot f(c)}{3}$.

4. Let $f : [1, \infty) \rightarrow \mathbb{R}$ be a differentiable positive function. Suppose that the improper integral $\int_1^\infty \frac{f'(x)}{f(x)} dx$ is convergent. Prove that $\lim_{x \rightarrow \infty} f(x)$ exists.

Solution:

We note that

$$\int_1^\infty \frac{f'(x)}{f(x)} dx = [\ln(f(x))]_1^\infty = \lim_{x \rightarrow \infty} [\ln(f(x)) - \ln(f(1))]$$

As $\int_1^\infty \frac{f'(x)}{f(x)} dx$ is convergent, then $\lim_{x \rightarrow \infty} [\ln(f(x)) - \ln(f(1))]$ exists.

Thus

$$\lim_{x \rightarrow \infty} \ln(f(x)) = \lim_{x \rightarrow \infty} [\ln(f(x)) - \ln(f(1)) + \ln(f(1))]$$

exists by AOL, i.e., there exists an $L \in \mathbb{R}$ such that $\lim_{x \rightarrow \infty} \ln(f(x)) = L$ and therefore

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} e^{\ln(f(x))} \stackrel{\text{AOL}}{=} e^L$$

5. Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a function that is integrable on $[0, b]$ for every $b > 0$.

Prove or disprove each of the following statements:

- (a) Consider the function defined by $F(x) = \int_0^x f(t) dt$ for every $x \in [0, 1]$. If F is differentiable at 0, then $F'(0) = f(0)$.

Solution:

False.

We choose

$$f(x) = \begin{cases} 1 & x = 0 \\ 0 & x \neq 0 \end{cases} \quad \forall x \in [0, 1]$$

Note that as f differs from a continuous function by only 1 point, then f is integrable on $[0, x]$ for every $x \in [0, 1]$ and

$$F(x) = \int_0^x f(t) dt = \int_0^x 0 dt = 0 \quad \forall x \in [0, 1]$$

Thus, F is differentiable at 0 and $F'(0) = 0 \neq f(0)$.

- (b) If $\int_0^1 f(x) dx = 2.3$, then there exists a $0 < c < 1$ for which $\int_0^c f(x) dx = c + 1$.

Solution:

True.

Consider the function defined by $F(x) = \int_0^x f(t) dt - x - 1$ for every $x \in [0, 1]$. As f is integrable on $[0, 1]$ then f is integrable on $[0, x]$ for every $x \in [0, 1]$ and therefore F is well defined. We also know that F is continuous by AOC. Note that $F(0) = \int_0^0 f(t) dt - 0 - 1 = -1$ and that $F(1) = \int_0^1 f(t) dt - 2 = 0.3$. Thus, by the IVT, there exists a point $0 < c < 1$ for which $F(c) = 0 \Rightarrow \int_0^c f(t) dt = c + 1$.

- (c) Consider the function defined by $F(x) = \int_x^1 f(t) dt$ for every $x \in [0, 1]$.

If f is continuous at 0, then $F'(0) = -f(0)$.

Solution:

True.

Consider the function defined by $G(x) = \int_0^x f(t) dt$ for every $x \in [0, 1]$. As f is continuous at $x_0 = 0$ then by the Fundamental theorem of Calculus, G is differentiable at $x_0 = 0$ and $G'(0) = f(0)$. Note that for every $x \in [0, 1]$ we have

$$F(x) = \int_x^1 f(t) dt = \int_0^1 f(t) dt - \int_0^x f(t) dt = \int_0^1 f(t) dt - G(x)$$

Thus, F is differentiable at $x_0 = 0$ by AOD and $F'(0) = -G'(0) = -f(0)$.

- (d) If the improper integral $\int_0^\infty f(x) dx$ converges absolutely, then the improper integral $\int_0^\infty \frac{f(x)}{x} dx$ is convergent.

Solution:

False.

We choose $f(x) = \frac{1}{1+x^2}$ for every $x \geq 0$. We have seen in class that the improper integral $\int_0^\infty f(x) dx$ satisfies

$$\int_0^\infty |f(x)| dx = \int_0^\infty \frac{dx}{1+x^2} = \frac{\pi}{2}$$

hence the integral converges absolutely. But

$$\int_0^1 \frac{f(x)}{x} dx = \int_0^1 \frac{dx}{x(1+x^2)} \geq \int_0^1 \frac{dx}{2x} = \frac{1}{2} \int_0^1 \frac{dx}{x} = \infty$$

and therefore $\int_0^1 \frac{f(x)}{x} dx = \infty$ and therefore $\int_0^\infty \frac{f(x)}{x} dx$ diverges.

6. For every $0 \leq n \in \mathbb{Z}$, consider the definite integral I_n defined by

$$I_n = \int_0^{\frac{\pi}{2}} \sin^n(x) dx$$

(a) Prove that the sequence $(I_n)_{n=0}^{\infty}$ is well-defined.

Solution:

For every $n \in \mathbb{N}$ the function $f(x) = \sin^n(x)$ for every $x \in [0, \frac{\pi}{2}]$ is continuous by AOC, and therefore integrable on $[0, \frac{\pi}{2}]$. It follows that the integral $I_n = \int_0^{\frac{\pi}{2}} \sin^n(x) dx$ exists, and consequently the sequence $(I_n)_{n=1}^{\infty}$ is well-defined. For $n = 0$ it holds that $f(x) = \sin^0(x) = 1$ for every $0 < x \leq \frac{\pi}{2}$, and therefore f has a removable discontinuity at $x = 0$. Consequently, f is integrable on $[0, \frac{\pi}{2}]$ and therefore I_0 is well-defined.

(b) Prove that the sequence $(I_n)_{n=0}^{\infty}$ satisfies the following recurrence formula

$$\begin{cases} I_n = \frac{n-1}{n} I_{n-2} & \forall n \geq 2 \\ I_0 = \frac{\pi}{2} \\ I_1 = 1 \end{cases}$$

Solution:

We first compute

$$I_0 = \int_0^{\frac{\pi}{2}} \sin^0(x) dx = \int_0^{\frac{\pi}{2}} 1 dx = \frac{\pi}{2}$$

and

$$I_1 = \int_0^{\frac{\pi}{2}} \sin(x) dx = [-\cos(x)]_0^{\frac{\pi}{2}} = -\cos\left(\frac{\pi}{2}\right) + \cos(0) = 1.$$

We now prove that $I_n = \frac{n-1}{n} I_{n-2}$ for every $n \geq 2$.

Let $n \geq 2$. We integrate by parts with $u' = \sin x$ and $v = \sin^{n-1}(x)$. Then

$$\begin{aligned} I_n &= \int_0^{\frac{\pi}{2}} \sin^n(x) dx = \int_0^{\frac{\pi}{2}} \sin(x) \cdot \sin^{n-1}(x) dx \stackrel{\text{IBP}}{=} [-\cos(x) \cdot \sin^{n-1}(x)]_0^{\frac{\pi}{2}} + (n-1) \int_0^{\frac{\pi}{2}} \cos(x) \cdot \sin^{n-2}(x) dx \\ &= (n-1) \int_0^{\frac{\pi}{2}} \cos(x) \cdot \sin^{n-2}(x) dx = (n-1) \int_0^{\frac{\pi}{2}} [1 - \sin^2(x)] \cdot \sin^{n-2}(x) dx \\ &= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2}(x) dx - (n-1) \int_0^{\frac{\pi}{2}} \sin^n(x) dx = (n-1) I_{n-2} - (n-1) I_n \end{aligned}$$

Thus, $I_n = (n-1) I_{n-2} - (n-1) I_n \Rightarrow n I_n = (n-1) I_{n-2} \Rightarrow I_n = \frac{n-1}{n} I_{n-2}$.

(c) Conclude that for every $n \in \mathbb{N}$ we have

$$I_{2n} = \frac{2n-1}{2n} \cdot \frac{2n-3}{2n-2} \cdots \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}$$

and

$$I_{2n+1} = \frac{2n}{2n+1} \cdot \frac{2n-2}{2n-1} \cdots \frac{4}{5} \cdot \frac{2}{3} \cdot 1$$

Solution:

We prove the claim by induction on n .

Base case: ($n = 1$) We already know that $I_0 = \frac{\pi}{2}$ and that $I_1 = 1$.

Induction step Let $n \in \mathbb{N}$ and assume the proposition holds for $2n-1$ and for $2n-2$. We prove that the proposition holds for $2n$ and for $2n+1$:

$$I_{2n} = \frac{2n-1}{2n} \cdot I_{2n-2} \stackrel{\text{induction hypothesis}}{=} \frac{2n-1}{2n} \cdot \frac{2n-3}{2n-2} \cdots \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}$$

and

$$I_{2n+1} = \frac{2n}{2n+1} I_{2n-1} \stackrel{\text{induction hypothesis}}{=} \frac{2n}{2n+1} \cdot \frac{2n-2}{2n-1} \cdots \frac{4}{5} \cdot \frac{2}{3} \cdot 1$$

(d) Conclude that $\lim_{n \rightarrow \infty} \frac{I_{2n+1}}{I_{2n}} = 1$.

Solution:

As $0 \leq \sin(x) \leq 1$ for every $x \in [0, \frac{\pi}{2}]$, then for every $n \in \mathbb{N}$ and for every $x \in [0, \frac{\pi}{2}]$ we have

$$I_{n+1} = \int_0^{\frac{\pi}{2}} \sin^{n+1}(x) dx \leq \int_0^{\frac{\pi}{2}} \sin^n(x) dx = I_n$$

It follows that the sequence $(I_n)_{n=1}^{\infty}$ is decreasing and therefore

$$I_{2n+1} \leq I_{2n} \Rightarrow \frac{I_{2n+1}}{I_{2n}} \leq 1, \quad \forall n \in \mathbb{N}$$

In addition, we know by item (c)

$$I_{2n+1} = \frac{2n}{2n+1} I_{2n-1} \geq \frac{2n}{2n+1} I_{2n} \Rightarrow \frac{I_{2n+1}}{I_{2n}} = \frac{2n}{2n+1} = \frac{1}{1 + \frac{1}{2n}}$$

We conclude that

$$\frac{1}{1 + \frac{1}{2n}} \leq \frac{I_{2n+1}}{I_{2n}} \leq 1, \quad \forall n \in \mathbb{N}$$

and by the squeeze rule, $\lim_{n \rightarrow \infty} \frac{I_{2n+1}}{I_{2n}} = 1$.

(e) Conclude the *Wallis* formula for π :

$$\frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdot \frac{6}{7} \cdots = \frac{\pi}{2}$$

Note: the left-hand side is defined by

$$\frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdot \frac{6}{7} \cdots = \lim_{n \rightarrow \infty} \left[\frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdot \frac{6}{7} \cdots \frac{2n}{2n-1} \cdot \frac{2n}{2n+1} \right]$$

Solution:

We have

$$\begin{aligned} \lim_{n \rightarrow \infty} \left[\frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdot \frac{6}{7} \cdots \frac{2n}{2n-1} \cdot \frac{2n}{2n+1} \right] &= \lim_{n \rightarrow \infty} \left[\frac{2 \cdot 4 \cdot 6 \cdots 2n}{3 \cdot 5 \cdot 7 \cdots (2n+1)} \cdot \frac{2 \cdot 4 \cdot 6 \cdots 2n}{1 \cdot 3 \cdot 5 \cdots (2n-1)} \right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{\left[\frac{2 \cdot 4 \cdot 6 \cdots 2n}{3 \cdot 5 \cdot 7 \cdots (2n+1)} \right]}{\left[\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots 2n} \right]} \right] = \lim_{n \rightarrow \infty} \left[\frac{I_{2n+1}}{\left[\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots 2n} \right] \cdot \frac{\pi}{2}} \right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{I_{2n+1}}{I_{2n}} \cdot \frac{\pi}{2} \right] = \frac{\pi}{2} \end{aligned}$$